Course Code/Course Name: Preparation of Placement Question Paper for Upcoming Drive

SET - 1

Q1. What is the value of $(a + b + c)^2 : (a^3 + b^3 + c^3)$ if $(ab + bc + ca) = 131$, $(a^2 + b^2 + c^2) = 138$ and $abc = 280$?
A. 15 : 43
B. 18 : 41
C. 20 : 49
D. 17 : 434
Answer: C

Q2. By which of the following is 19541742 divisible?
I. 11
II. 9
III. 12

A. Both I and III
B. Only III
C. Both II and III
D. Only I
Answer: D

Q3. Simplify

$$\frac{0.7 \times 0.7 \times 0.7 + 0.3 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 3}{0.7 \times 0.7 + 0.3 \times 0.3 + 0.42}$$

A. 1
B. 1/3
C. -⅓
D. 3
Answer: A

Q4. In an examination, 62% candidates passed in physics and 60% candidates passed in Mathematics. If 37% candidates passed in both these subjects, what percent of the candidates failed in both the subjects?
A. 5%
B. 20%
C. 25%
D. 15%
Answer: D

Q5. If price of petrol increases by 35% and Rajesh intends to spend only an additional 25% on petrol, by how much % will he reduce the quantity of petrol purchased?(Approx)
A. 14%
B. 9%
C. 11%
D. 13%
Answer: C

Q6. If 20% apples from a cart are rotten and the remaining apples are sold at 20% discount, the seller gets 92% profit on the cost price of the whole cart. By what percent above the cost price has the seller marked up the apples?
A. 180%
B. 250%
C. 200%
D. 280%
Answer: C

Q7. Selling an item at 5/6th of its marked price results in a loss of 10%. What is the % of profit/loss, if a discount of 5% is offered on the marked price?
A. 2.6% loss
B. 2.6% profit
C. 3% profit
D. 3% loss
Answer: B

Q8. The simple interest earned on a sum of Rs. 3650 at some rate after 4 years is Rs. 584. Find the rate o interest (in p.a.)
A. 3.5%
B. 4.5%
C. 3%
D. 4%
Answer: D

Q9. If a : b = 3 : 4, b : c = 2 : 3, c : d = 1 : 2, find the value of ((a+b+c+d)/(b+d))^½
A. 5 : 4
B. 4 : 5
C. 1 : 1
D. 2 : 3
Answer: A

Q10. Which of the following statements is true?
I) 27840 is divisible by 12
II) 7329753 is divisible by 9
A. Statement I is true, but II is not
B. Statement I is not true, but II is true
C. Both the statements are true
D. Both the statements are not true
Answer: C

Q11. What is the value of the following expression?

$$\left(3\frac{3}{4}\text{ of }5\frac{1}{7}\div\frac{9}{7}\right)\times 3\frac{1}{3}\div\left(1\frac{3}{4}\times 2\frac{2}{3}\text{ of}\frac{6}{7}\div\frac{4}{5}\right)$$

A. 10
B. 12
C. 8
D. 9
Answer: A

Q12. A sum amounts to Rs. 16,000 at the end of three years and to Rs. 25,600 at the end of six years under compound interest (interest being compounded annually). Find the simple interest earned if the same sum is invested at 15% p.a. for two years.
A. Rs. 3,000
B. Rs. 4,000
C. Rs. 3,500
D. Rs. 4,500
Answer: A

Q13. A flask is filled with 20% acid. 49 ml of that solution is taken out and is replaced with 40% acid to make it a solution of 27% acid. Find the initial volume of the solution in the flask.
A. 140 ml
B. 175 ml
C. 70 ml
D. 105 ml
Answer: A

Q14. Disha and Patni entered into a partnership. Disha invested as much money as was earned by Patni a share from the profit. If the profit was Rs. 18,000 and the money invested by Disha was Rs. 3,000 more than Patni, What is the investment made by Disha if it is less than Rs. 5,000?
A. Rs. 4,500
B. Rs. 7,500
C. Rs. 6,000
D. Rs. 5,500
Answer: A

Q15. If the 9-digit number 807x6y9z8 is divisible by 99, then the value of (x + y + z)^½ is __
A. 4
B. $3\sqrt{3}$
C. 6
D. $\sqrt{5}$
Answer: A

Q16. Rahul and Sachin can complete a work in 60 days and 45 days respectively. Rahul starts working with 120% of his capacity and Sachin with 80% of his capacity. In how many days will they complete 75% of the work if they work together?
A. 25.71
B. 19.85
C. 17.64
D. 16.24
Answer: B

Q17. The ratio of the speed of a bus and a car is 2:5. The car covers a distance of 280 km in some time and the bus covers a distance of d km less in the same time. What is the distance covered by the bus?
A. 132 km
B. 168 km
C. 112 km
D. 144 km
Answer: C

Q18. A boat travels 12 km downstream and 6 km upstream in 3 hours. The same boat takes fifty percent extra time to cover 10 km downstream and 16 upstream. If the same boat travels 20 km downstream and z km upstream in 4 hours, find z.
A. 8 km
B. 4.5 km
C. 7.5 km
D. 9 km
Answer: B

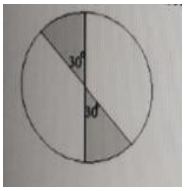
Q19. Ranjeet's present age is half of his brother Manjeet's age after 10 years. After 5 years, the ratio of ages of Manjeet and Ranjeet will be 7 : 5. Find Ranjeet's age 2 years ago.
A. 25 yrs
B. 20 yrs
C. 15 yrs
D. 18 yrs
Answer: D

Q20. A plane is cutting a cone parallel to the base in such a way that the radius of the new cone is half of the radius of the original cone. Find the ratio of the volume of the original cone to that of the volume of the portion of the cone left after being cut by the plane (the part other than the new cone).
A. 8 : 7
B. 9 : 7
C. 7 : 6
D. 8 : 5
Answer: A

Q21. Find the geometric mean of the data 5, 75 and 9.

A. 15
B. 5
C. 45
D. 3
Answer: A

Q22. In a circular ground of radius 28m, the shaded portion area in the circle represents forest area while the remaining part of the circle is available for construction purposes. Find the area reserved for construction purpose(approx).



A. 1764 m^2
B. 3120 m^2
C. 1850 m^2
D. 2053 m^2
Answer: D

Q23. Find the difference between arithmetic mean and median of the first seven consecutive natural numbers.
A. 1
B. 3.5
C. 2
D. 0
Answer: D

Q24. Find the quartile deviation of the observations 16, 2, 8, 24, 4, 32, 18.
A. 8
B. 12
C. 6
D. 10
Answer: D

Q25. The standard deviation of a series of 'n' observations is σ. If each observation is multiplied by 12, the respective ratio of the standard deviation and the variance is:
A. 1 : 9σ
B. 1 : 12σ
C. 3σ : 1
D. σ : 1
Answer:

Q26. If the arithmetic mean of 23, 15, 19, p, 12, 7, is 16 while the arithmetic mean of 37, 28, 53, q, 42 is 4[illegible], then find (p+q)/(q-p)

A. 2.6

B. 2.2

C. 3.5

D. 3.2

Answer: A

Q27. Which is the WRONG term in the following series?

LAT, OES, QIR, SOQ, UUP

A. QIR

B. LAT

C. UUP

D. SOQ

Answer: A

Q28. The table shows the total number of 10th class students in 5 different schools [P, Q, R, S and T] and the ratio of the number of students in Art, Commerce and Science respectively.

The students study only the above subjects in the class of each school.

School	Total Students	Arts : Commerce : Science
P	240	1 : 2 : 3
Q	450	7 : 1 : 1
R	360	5 : 4 : 3
S	180	2 : 5 : 3
T	330	6 : 4 : 1

What is the total number of Science students in school P, Q and R together?

A. 290

B. 220

C. 260

D. 240

Answer: C

Q29. In each of the five pairs of letter-clusters, the letters in the second term are a rearranged/transformed form of the letters in the first term in a particular pattern. In which two pairs, has the transformation been done in the same way?

A. PLANT : PLZMG

B. MANGO : MAMTL

C. BLACK : BLBDP

D. TABLE : ATLBE

E. CHEER : HCREE

A. A and B

B. D and A

C. B and C

D. C and D

Answer: A

Q30. Given below is a question followed by two statements, I and II each containing some information. Decide which of the statement(s) is/are sufficient to answer the question.

Five persons - A, B, C, D and E - sit around a circular table, facing the centre not necessarily in that order. Who sits second to the right of D?

I. A sits to the immediate right of E, who is not a neighbour of D.

II. B is to the immediate left of D.

A. If only statement II is sufficient to answer the question

B. If the data in both the statements I and II together are necessary to answer the question.

C. If only statement I is sufficient to answer the question.

D. If the data either in statement I alone or in statement II alone is sufficient to answer the question.

Answer: B

Q31. Given below is a question followed by two statements, I and II each containing some information. Decide which of the statement(s) is/are sufficient to answer the question.

Six persons P, Q, R, S, T and U have different scores in a classroom test, who scored the 2nd highest marks?

I. P scores better than T and Q, who scores better than S. S scores better than U. R scores better than P.

II. S scores better than at least one person, P scores better than S and T.

A. The data is statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question

B. The data given in both statements I and II together is not sufficient to answer the question

C. The data either in statement I alone or in statement II alone is sufficient to answer the question

Q. The data is statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.

Answer: A

Q32. In a certain code, if:

I. J @ K means J is K's daughter.

II. J % K means J is K's wife.

III. J + K means K is J's father.

IV. J ÷ K means K is J's mother.

F @ R % I ÷ E % N

How is R related to N in the given expression?

A. Daughter-in-law

B. Mother-in-law

C. Son-in-law

D. Mother

Answer: A

Q33. The annual salary of a person consists of two parts - basic salary and allowance. The allowance is always calculated as a fraction of the basic salary. Every year, his basic salary increases by Rs.5000 and his

allowance, as a percentage of his basic salary, increases by 5 percentage points. The present basic salary of the person is Rs. 30,000 and his present allowance is 20% of his present basic salary. After how many months from now will the Annual salary of the person be Rs. 60,750?

A. 33
B. 30
C. 36
D. 24

Answer: C

Q34. In a certain code, if:
K @ L means K is L's father.
K # L means K is daughter of
L. K % L means L is mother of
K. K + L means K is grandson
of L. K × L means L only child
of K.
K © L means K is L's sister.
R + A × E @ F # G © T
Which of the following is a CORRECT conclusion drawn from the expression?
A. F is daughter of A
B. R is father of T.
C. E is mother of T.
D. R is son of G.

Answer: D

Q35. The given question consists of six statements followed by 4 options consisting of three statements put together in a specific order. Choose the option, which indicates a valid argument that is, when the third statement is a conclusion drawn from the preceding two statements.

a. Magazines are published weekly.
b. Some magazines are not newspapers.
c. Newspapers are published daily.
d. Magazine is not published daily.
e. Magazines are not newspapers.
f. Some newspapers are not published weekly.

A. b f e
B. a f c
C. d c f
D. a c e

Answer: D

Q36. Three statements are given followed by two conclusions numbered I and II. You have to consider these statements to be TRUE, even if they seem to be at variance with commonly known facts. Decide which of the given conclusions logically follows from the given statement.

Statements:
Some books are cars.
No bus is a car.
Some cars are motors.

Conclusions:
I. Some motors are not buses.
II. Some books are buses.

A. If either I or II follows
B. If neither I or II follows
C. If only II follows
D. If only I follows

Answer: B

Q37. A statement is followed by two assumptions numbered I and II. Consider the statement and decide which of the given assumptions is/ are implicit in the statement.

Statement:
Country X will talk on every issue and try to resolve every conflict with its neighbouring country.

Assumptions:
I. Country X wants a peaceful relation with its neighbouring country.
II. Country X will appear weaker than the neighbouring country if it tries to solve the matter by talking.

A. Neither I nor II is implicit.
B. Only II is implicit.
C. Only I is implicit.
D. Both I and II are implicit.

Answer: C

Q38. A statement is followed by two arguments numbered I and II. Decide which of the argument(s) is/ are strong (important and directed related to the question) and which is/ are weak (not important and not related to the central message of the question).

Statement:
Should young children be exposed to virtual gaming?

Arguments:
I. Yes. It will prepare them for leadership.
II. No. Young minds cannot take violence.

A. Both I and II are strong.
B. Only II is strong.
C. Only I is strong.
D. Neither I nor II is strong.

Answer: A

Q39. Two statements are given followed by two conclusions numbered I and II. You have to consider these statements to be TRUE even if they seem at variance with commonly known facts. Decide which of the conclusions logically follows from the given statement.

Statements:
All runs are overs.
Some overs are bats.

Conclusions:
I. All bats are runs.
II. No bat is a run.

A. If either I or II follows.
B. If neither I nor II follows.
C. If only I follows.
D. If only II follows.
Answer: B

Q40. Following are the selection criteria for any student to get admission in "Central University of India for
any undergraduate program.

a. Must have completed high school with at least 8.0 CGPA.
b. Must have been at least 17 years 8 months old on Jan 1st 2021.
c. Must have scored more than 80 percentage in university entrance test.
d. If students fulfill all the criteria except a, but have more than 6.5 CGPA in high school, his case will be sent to the Sr. Admission Officer.
e. If students fulfill all the criteria except b, but have more than 90 percentage in the university entrance test, his case will be sent to the Head of the Department.

Tanmay has passed high school with 9.2 CGPA. He has scored 92 percent in university entrance test and he was born on 27th April 2004.

A. His case will be sent to the Sr Admission Officer.
B. His case will be sent to the Head of the Department.
C. He will be selected.
D. He will not be selected.

Answer: B

Q41. Rohan is selecting pizza from a pizzeria. Following are the criteria of selection by him for the pizza.
a. It must be a non-veg pizza.
b. It must cost less than INR 450.
c. It must be a hand toasted pizza.
d. If any pizza is satisfying all the criteria given above except c, but it is a cheese burst pizza, then he needs permission from his uncle.
e. If any pizza is satisfying all the criteria given above except b, but provides mushroom and corn for free, then he needs permission from his mother.

"Pizza on the spot" house is providing a new hand toasted non-veg pizza which costs Rs.550. It provides free toppings of onion, capsicum, corn, paneer and tomato.

A. He will need to take permission from his uncle.
B. It will not be selected.
C. He will need to take permission from his mother.
D. It will be selected.

Answer: B

Q42. Five persons A,B,C,D, and E are comparing the number of books which they have bought recently. A bought more number of books than C and E, but not the highest. B bought more number of books than exactly one person. Then which of the following bought the highest numbers of books?

A. C
B. E
C. D
D. B

Answer: Option C

Q43. Rajpal is going to buy a laptop. Following are the criteria of selection of the laptop given by his father.
a. Must be < Rs.38,000.
b. Must have RAm equal to or more than 8 GB.
c. Must have intel processor of version i3 or above.

d. If any laptop is satisfying all of the above criteria except a, it will need his dad's permission if it is less than
Rs.45,000.
e. If any laptop is satisfying all of the above criteria except b, it will need his mom's permission if it has storage
of more than 512 GB.

ATC58 is a new laptop in the market that costs Rs.36,000and having 16 GB RAM. It consists of Pentium 8 processor and pre-installed Windows 10.

Which of the following options below is CORRECT for the laptop given above?

A. He needs his dad's permission to buy it.
B. He needs his mom's permission to buy it.
C. He will buy the laptop.
D. He will not buy the laptop.

Answer: D

Q44. I, J, K, L, M and N have their birthdays not in the same order but in different months such as January, February, April, July, August and December.

Two people have their birthdays between N and J's birthday. M's birthday is not in the second half of the year. The birthday of N is either in January or August. K's birthday falls in the month having the minimum number of days. At the most, 3 people have birthdays between N and I's birthday.

Whose birthday falls exactly between N and I's birthday?
A. Person I
B. J, birthday in July
C. Person J
D. The one whose birthday is in April.

Answer: D

Q45. Eight persons A,B,C,D,E,F,G and H are sitting around a square table . All of them are facing towards the centre. Only one person sits on each of the sides and only one person sits on each of the corners. Four of them sit on the sides and four of them sit on the corners. C is sitting 3rd to the left of B, who sits on a side. D is sitting opposite to H, who is sitting 3rd to the right of B. A is sitting 2nd to the right of G. G is not sitting opposite to either E or F, F is not an immediate neighbor of H.
Who among the following is sitting opposite to A?
A. C
B. B
C. F
D. E

Answer: Option D

Q46. Six friends L, M, N, O, P and Q are sitting around a circular table and facing outside. Q is sitting fourth to the left of M, who is sitting to the immediate left of N. P is sitting opposite to N and and adjacent to O. O is not sitting adjacent to Q.

Which of the following statements is TRUE?

A. L is adjacent to P
B. M is between O and P
C. P is sitting opposite to Q
D. There are 3 friends sitting between M and L.

Answer: A

Q47. In an examination 38% of students fail in English and 61% pass in Hindi and 23% fail in both. Find the actual failure percentage?

A)54% B)61% C)46% D)70%

Answer: A

Q48. Three candidates, Ajay, Bijoy & Chandu contested an election and received 1800, 3300 and votes 3900 respectively. What percent of the total votes did A get?

A)20% B)40% C)50% D)90%

Answer: A

Q49. The total population of a village increased from 1,80,00 to 22, 500 in a decade. The average percentage increase of population per year of that village is:

A)2.37% B)2.5% C)3. 6% D)6.75%

Answer: B

Q50. If the cost price is 25% of selling price. Then what is the profit percent.

A)150% B)200% C)300% D)350%

Answer: C